웹 개발자 부트캠프 과정

CodingOn

Java

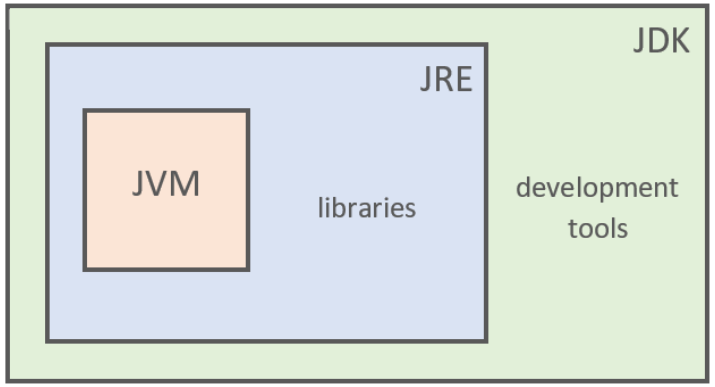
Java

- 썬 마이크로시스템즈에서 1995년에 처음 발표한 프로그래밍 언어
- 웹, 안드로이드 애플리케이션, 데스크톱 애플리케이션 등을 개발하는데 사용 가능
- 오라클(https://www.oracle.com/kr/)이 라이선스를 가지고 있음
- 오라클에서 JDK를 배포하여 자바로 프로그램을 개발할 수 있도록 지원함

- 특징
  - 모든 운영체제(윈도우, 맥, 리눅스)에서 실행 가능
  - 객체 지향 프로그래밍 (OOP, Object Oriented Programming) 언어
  - 메모리(RAM)을 자동으로 관리하여 개발자가 메모리 관리를 신경쓰지 않고 개발할 수 있음
  - 오픈 소스 라이브러리가 다양

개발환경 구축

용어 정리



- JDK (Java Development Kit)
  - 자바로 개발하는데 사용되는 SDK(소프트웨어 개발 키트)
- JRE (Java Runtime Environment)
  - 자바 실행 환경으로 자바 프로그램을 시킬 때 필요한 라이브러리 API를 묶어 배포되는 패키지
  - JDK 11 부터 JDK에 기본적으로 포함됨
- JVM (Java Virtual Machine)
  - 자바 가상 머신으로 자바를 돌리는 프로그램
  - 자바로 작성된 모든 프로그램은 JVM 에서만 실행 가능!
  - JRE에 포함됨. 따라서 컴퓨터 운영체제에 맞는 JRE가 설치되어 있다면 JVM 도 설치된 것
  - JVM은 모든 플랫폼에서 제약 없이 동작함. 즉, OS에 종속적이지 않다!

JVM, 자바 가상 머신

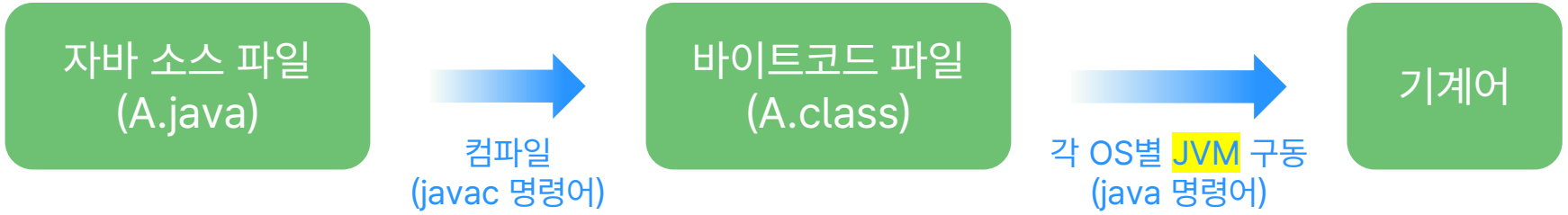
- Java는 플랫폼에 독립적인 언어

- 만약 플랫폼에 종속적이라면?
  - Windows에서 빌드한 프로그램을 Linux 혹은 macOS에서 실행 불가능
  - 빌드(build)란? 소스 코드를 실행 가능한 소프트웨어 제품으로 변환하는 과정

- 하지만 Java는?
  - JVM이 설치되어 있으므로 실행 가능
  - 각 운영체제 별로 JVM이 있어 JVM이 설치된 어떤 플랫폼에서도 실행 가능

JVM, 자바 가상 머신

- JDK를 설치하면 자바 언어로 작성한 소스 파일을 만들고 컴파일 가능
  - 컴파일: 고급 언어로 작성된 파일(source 파일)을 컴퓨터가 이해할 수 있는 기계어(0, 1)로 변환
  - 고급 언어: 사람이 사용하는 언어 (ex. Java, C, C++, C#, Python 등)



어떤 운영체제(Windows, macOS, Linux 등)더라도
동일한 소스 파일을 컴파일하면
모두 동일한 바이트코드 파일이 생성

java 명령어는 JVM을 구동시켜 바이트코드 파일을 기계어로 번역
운영체제에서 이해하는 기계어가 다르기 때문에
서로 다른 JDK를 설치했던 것!

JDK 설치

- https://www.oracle.com/java/technologies/downloads/#java17

Java 19 and Java 17 available now

Java 17 LTS is the latest long-term support release for the Java SE platform. JDK 19 and JDK 17 binaries are free to use in production and free to redistribute, at no cost, under the Oracle No-Fee Terms and Conditions.

Learn about Java SE Subscription

JDK 19 will receive updates under these terms, until March 2023 when it will be superseded by JDK 20.

JDK 17 will receive updates under these terms, until at least September 2024.

Java 19 **Java 17**

Java SE Development Kit 17.0.5 downloads

Thank you for downloading this release of the Java™ Platform, Standard Edition Development Kit (JDK™). The JDK is a development environment for building applications and components using the Java programming language.

The JDK includes tools for developing and testing programs written in the Java programming language and running on the Java platform.

Linux **macOS** **Windows** 각자 운영체제에 맞는 Tab 선택

Product/file description	File size	Download
x64 Compressed Archive	171.88 MB	https://download.oracle.com/java/17/latest/jdk-17_windows-x64_bin.zip (sha256)
x64 Installer	152.85 MB	https://download.oracle.com/java/17/latest/jdk-17_windows-x64_bin.exe (sha256)
x64 MSI Installer	151.73 MB	https://download.oracle.com/java/17/latest/jdk-17_windows-x64_bin.msi (sha256)

설치 확인

설치 확인 명령어 java -version

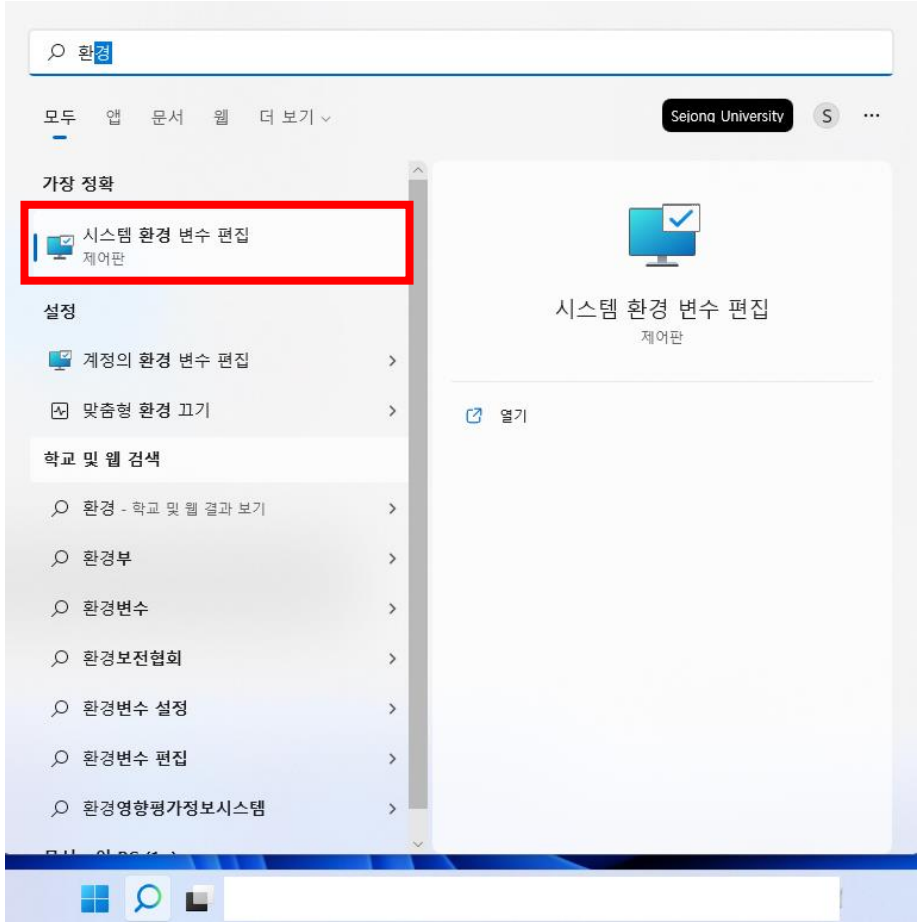
Windows - CMD에서

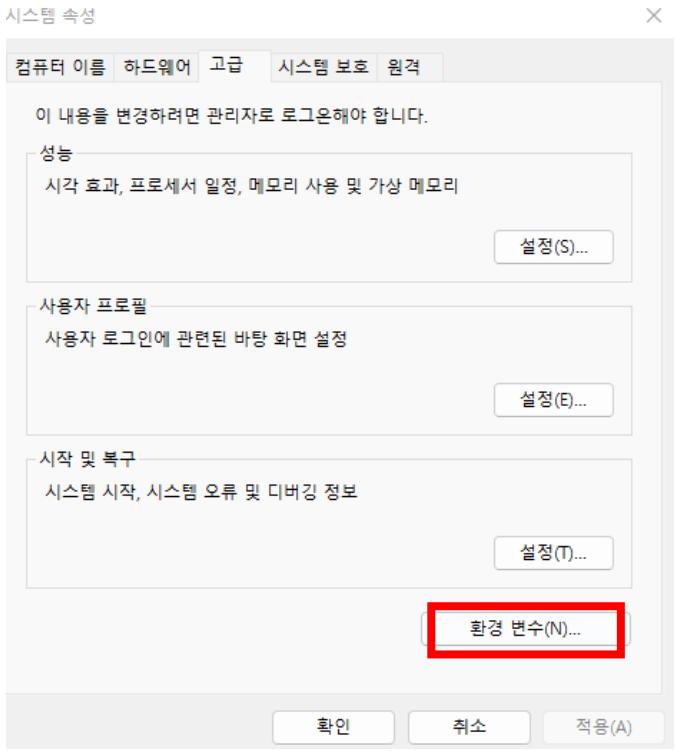
```
C:₩Users₩Lily>java -version
java version "17.0.5" 2022-10-18 LTS
Java(TM) SE Runtime Environment (build 17.0.5+9-LTS-191)
Java HotSpot(TM) 64-Bit Server VM (build 17.0.5+9-LTS-191, mixed mode, sharing)
```

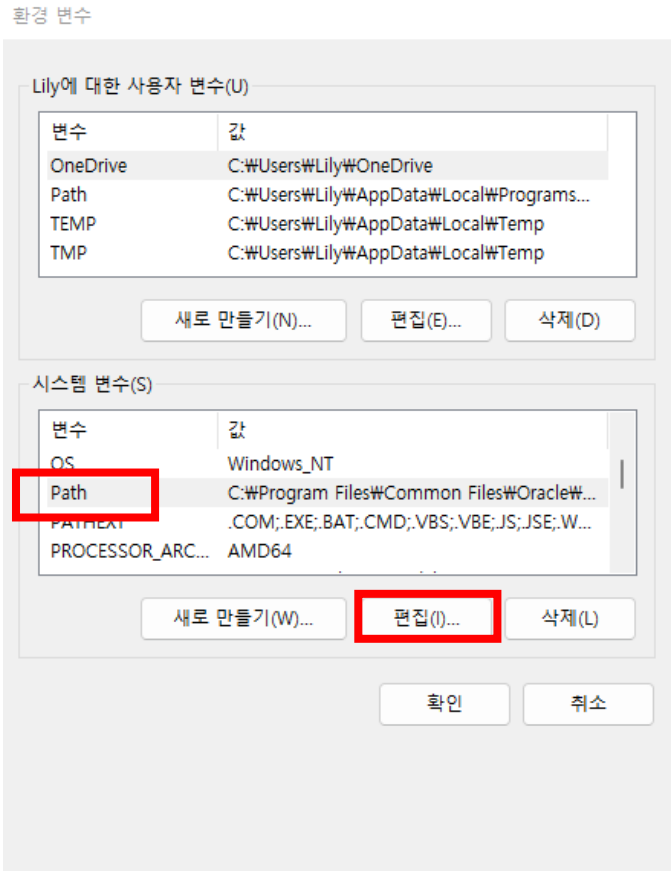
Mac OS - Terminal에서

```
-MacBookPro ~ % java -version
```

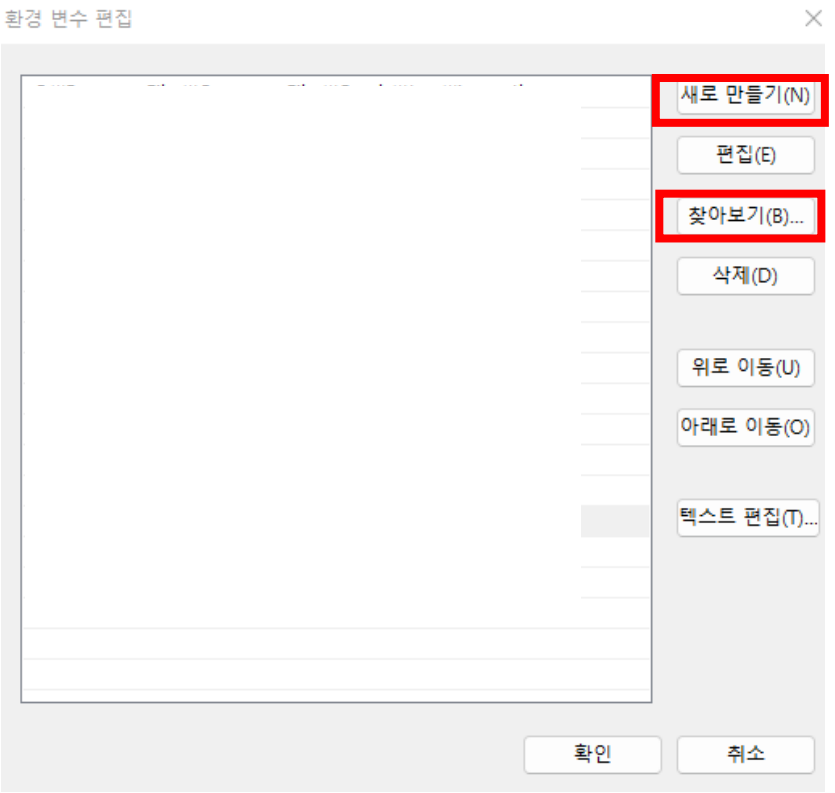
(참고) Windows 환경변수 설정

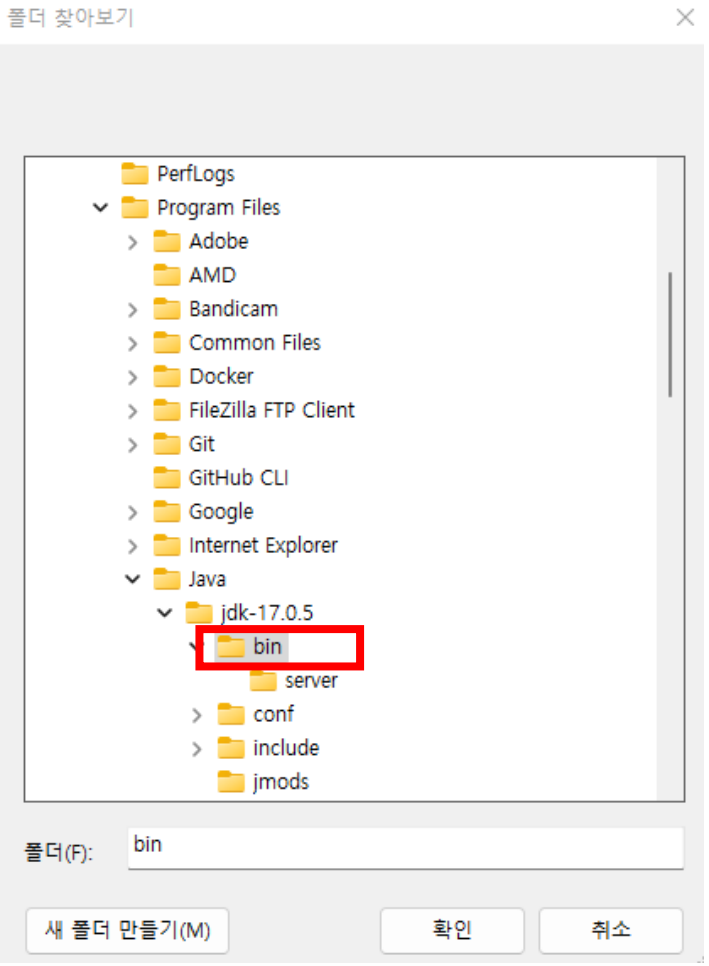


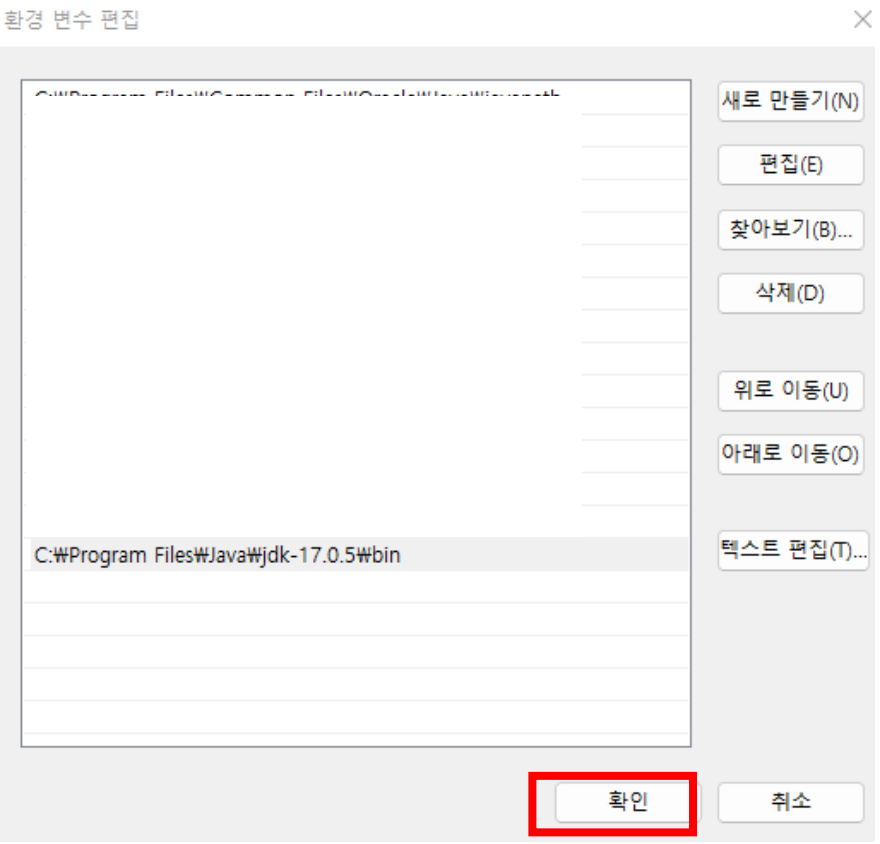




(참고) Windows 환경변수 설정







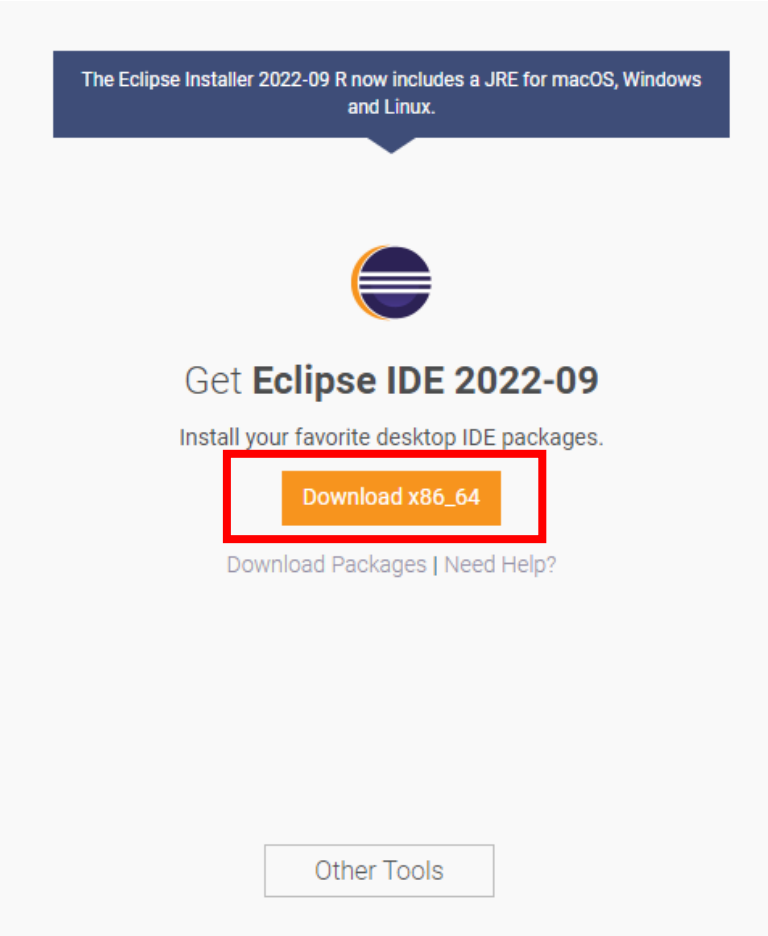
Eclipse 설치

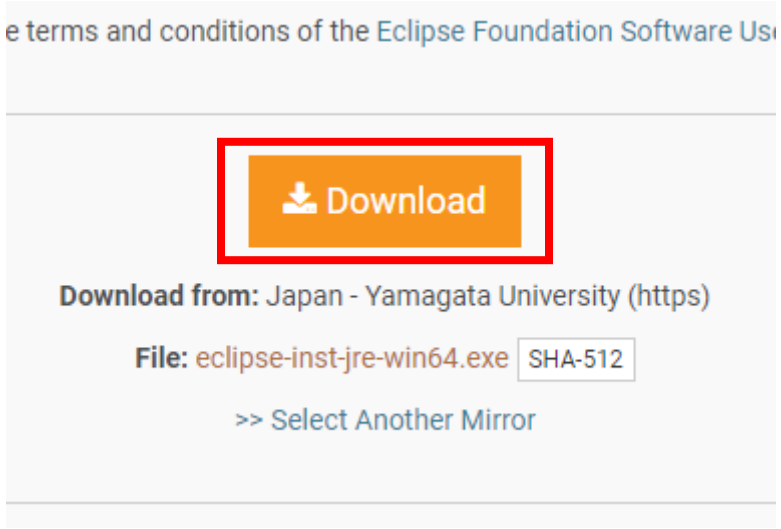
Eclipse 이클립스

- 현업에서 많이 사용하는
  통합 개발 환경 (IDE, Integrated Development Environment)
- 무료!
- 여러 프로젝트 동시 관리 가능

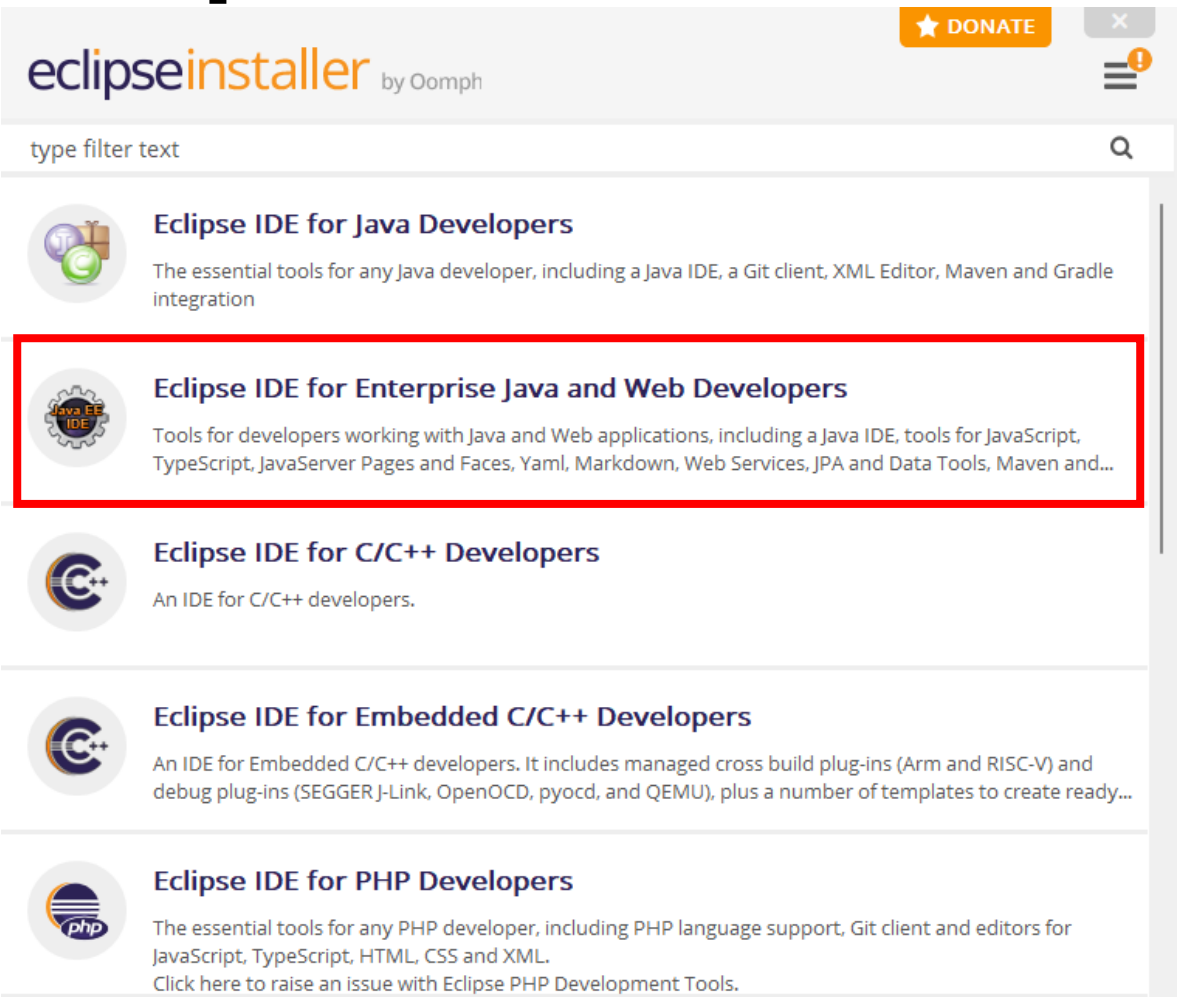
eclipse 설치

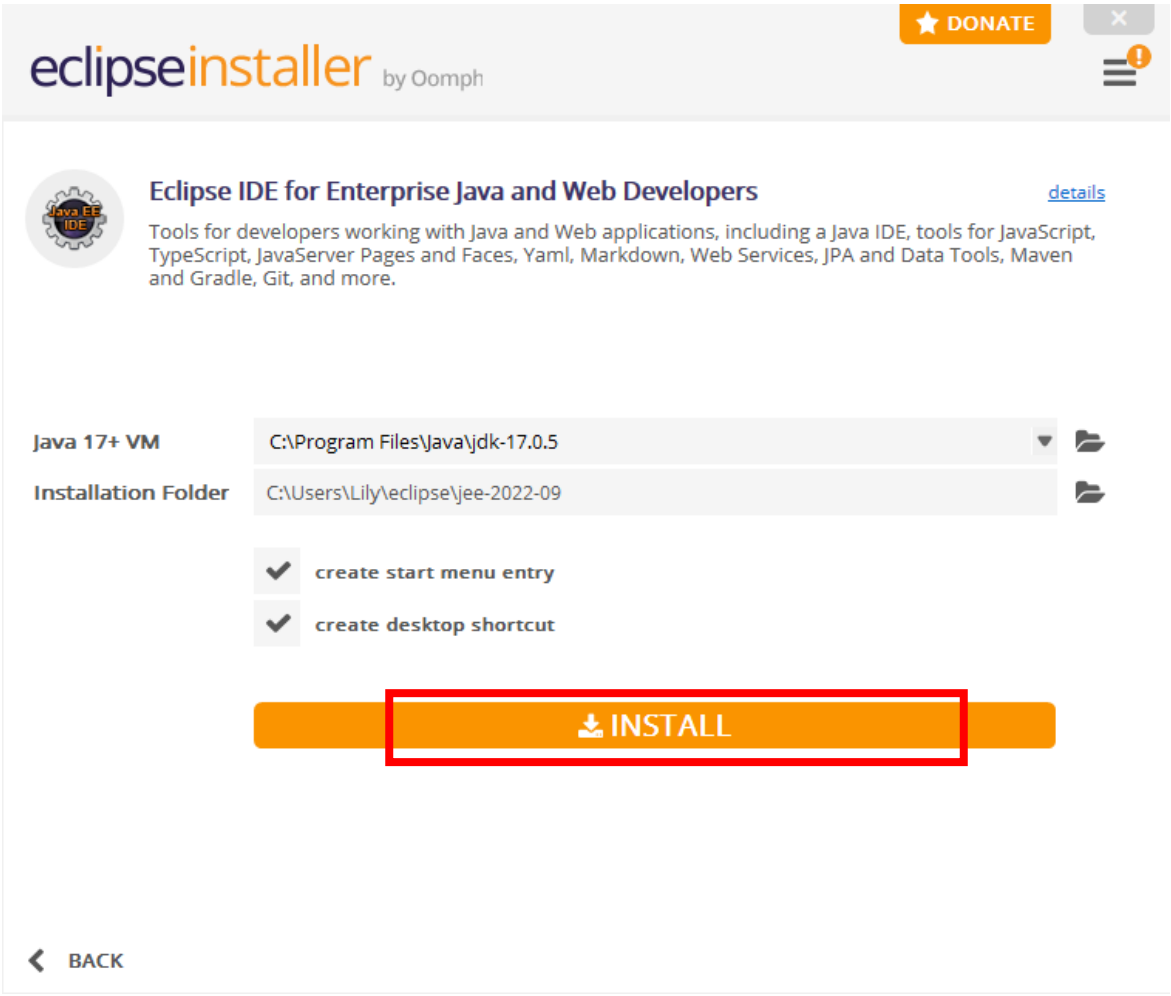
- https://www.eclipse.org/downloads/



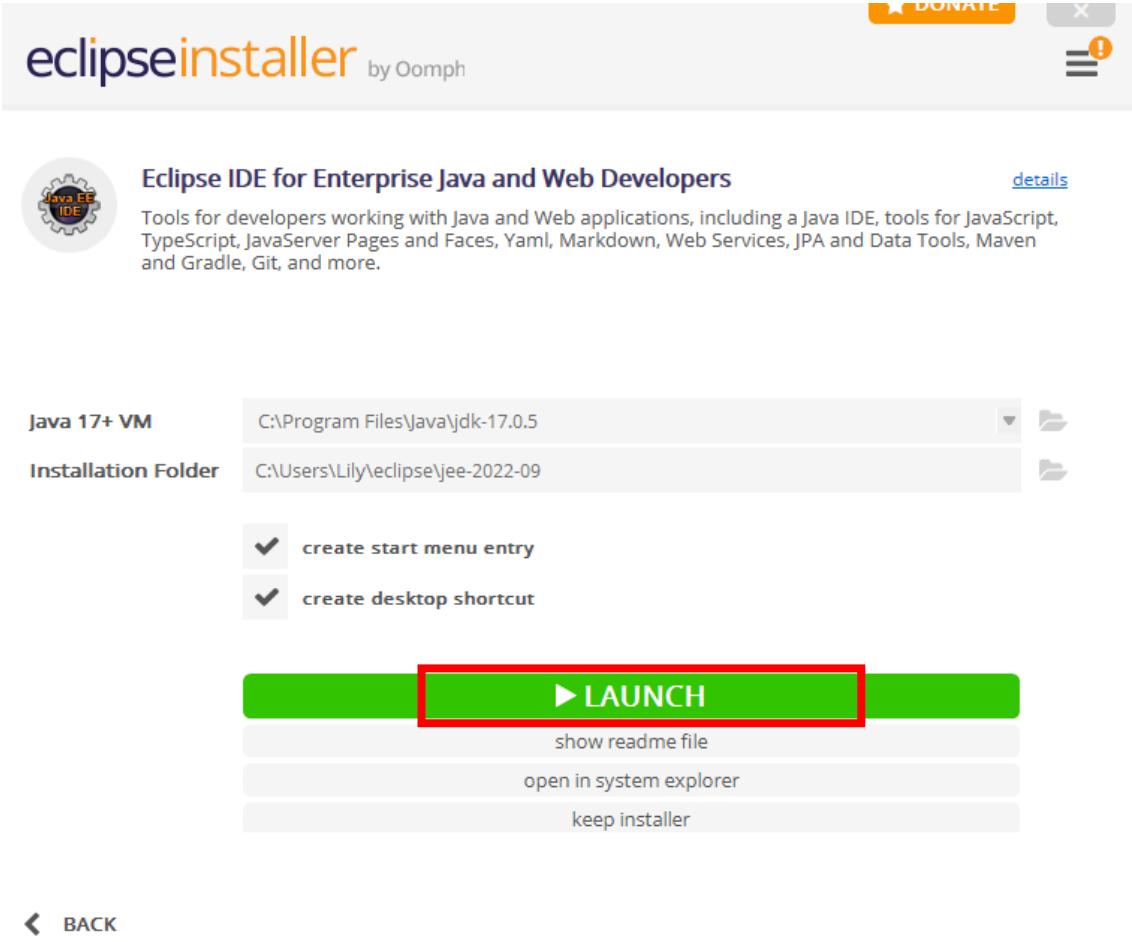


eclipse 설치

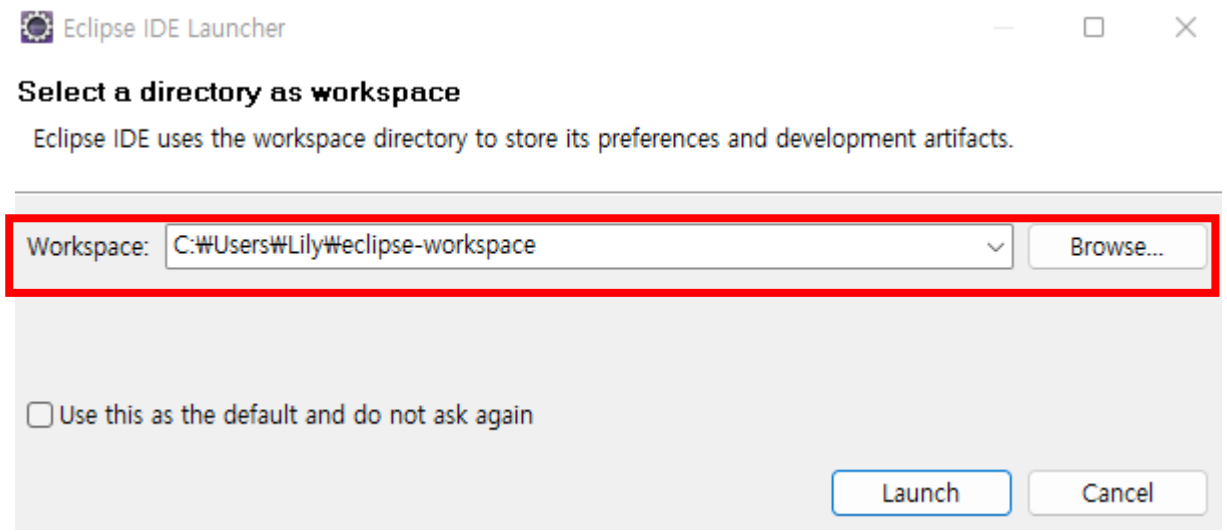

DONATE
eclipseinstaller by Oomph
type filter text
Eclipse IDE for Java Developers
The essential tools for any Java developer, including a Java IDE, a Git client, XML Editor, Maven and Gradle integration
Eclipse IDE for Enterprise Java and Web Developers
Tools for developers working with Java and Web applications, including a Java IDE, tools for JavaScript, TypeScript, JavaServer Pages and Faces, Yaml, Markdown, Web Services, JPA and Data Tools, Maven and...
Eclipse IDE for C/C++ Developers
An IDE for C/C++ developers.
Eclipse IDE for Embedded C/C++ Developers
An IDE for Embedded C/C++ developers. It includes managed cross build plug-ins (Arm and RISC-V) and debug plug-ins (SEGGER J-Link, OpenOCD, pyocd, and QEMU), plus a number of templates to create ready...
Eclipse IDE for PHP Developers
The essential tools for any PHP developer, including PHP language support, Git client and editors for JavaScript, TypeScript, HTML, CSS and XML.
Click here to raise an issue with Eclipse PHP Development Tools.


DONATE
eclipseinstaller by Oomph
Eclipse IDE for Enterprise Java and Web Developers
details
Tools for developers working with Java and Web applications, including a Java IDE, tools for JavaScript, TypeScript, JavaServer Pages and Faces, Yaml, Markdown, Web Services, JPA and Data Tools, Maven and Gradle, Git, and more.
Java 17+ VM
C:\Program Files\Java\jdk-17.0.5
Installation Folder
C:\Users\Lily\eclipse\jee-2022-09
create start menu entry
create desktop shortcut
INSTALL
BACK

eclipse 설치

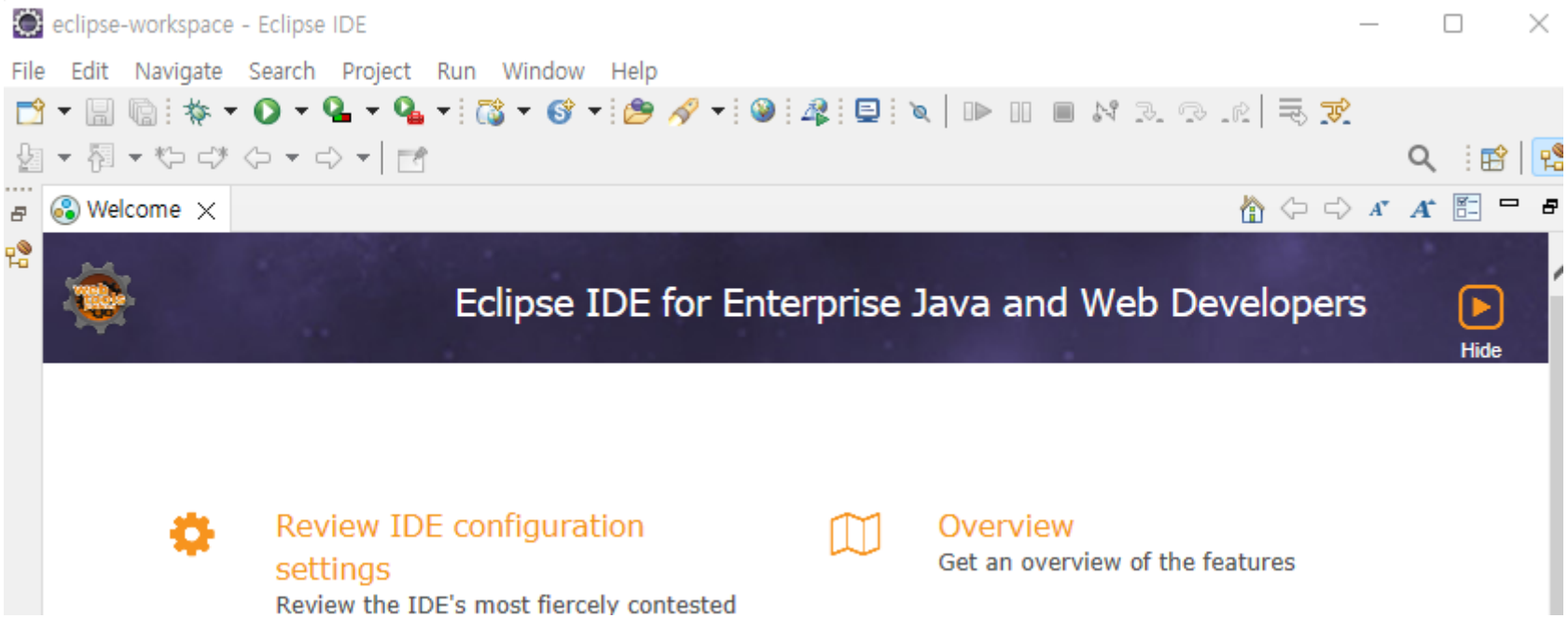


Workspace 가 위치할 곳 설정

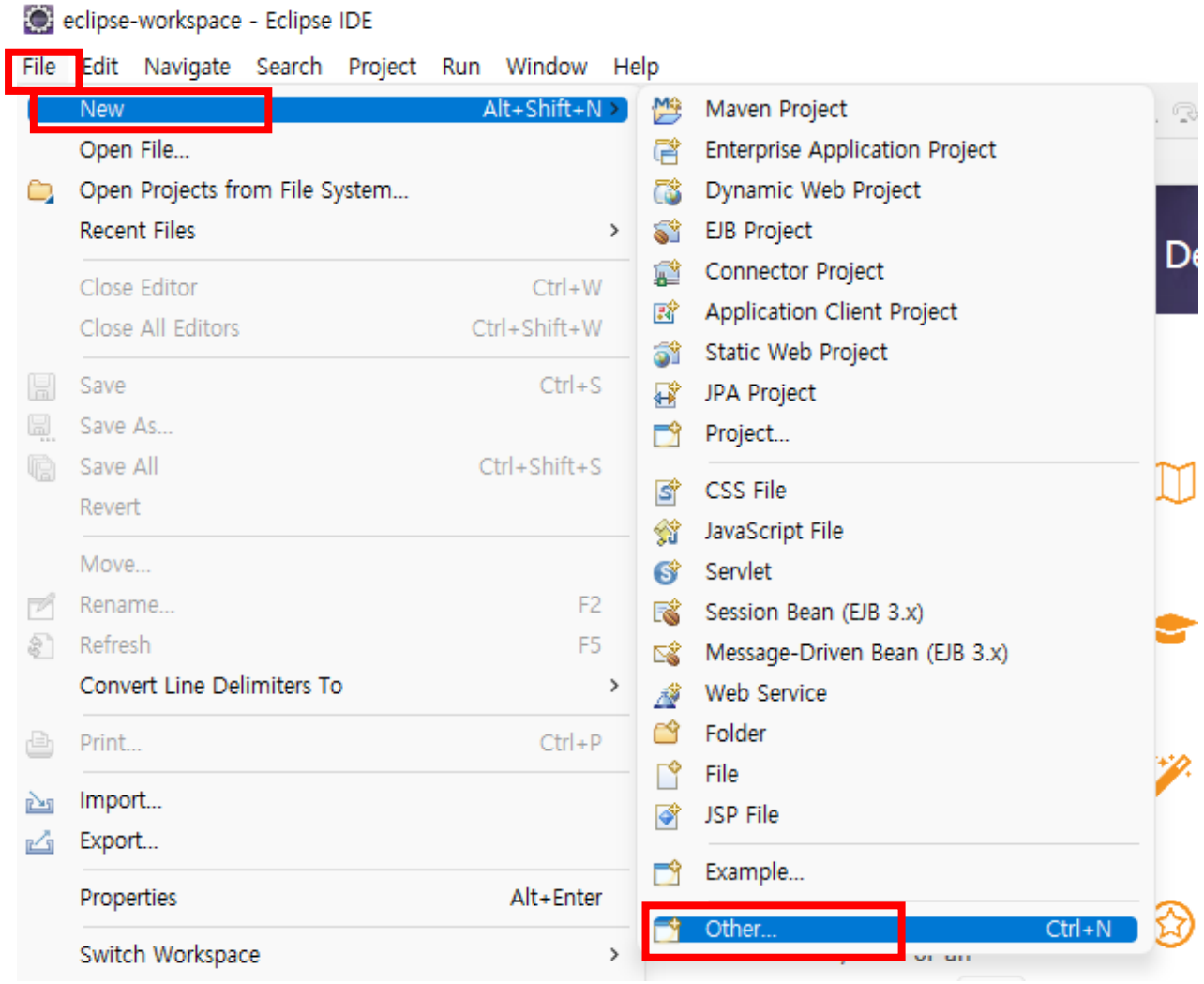


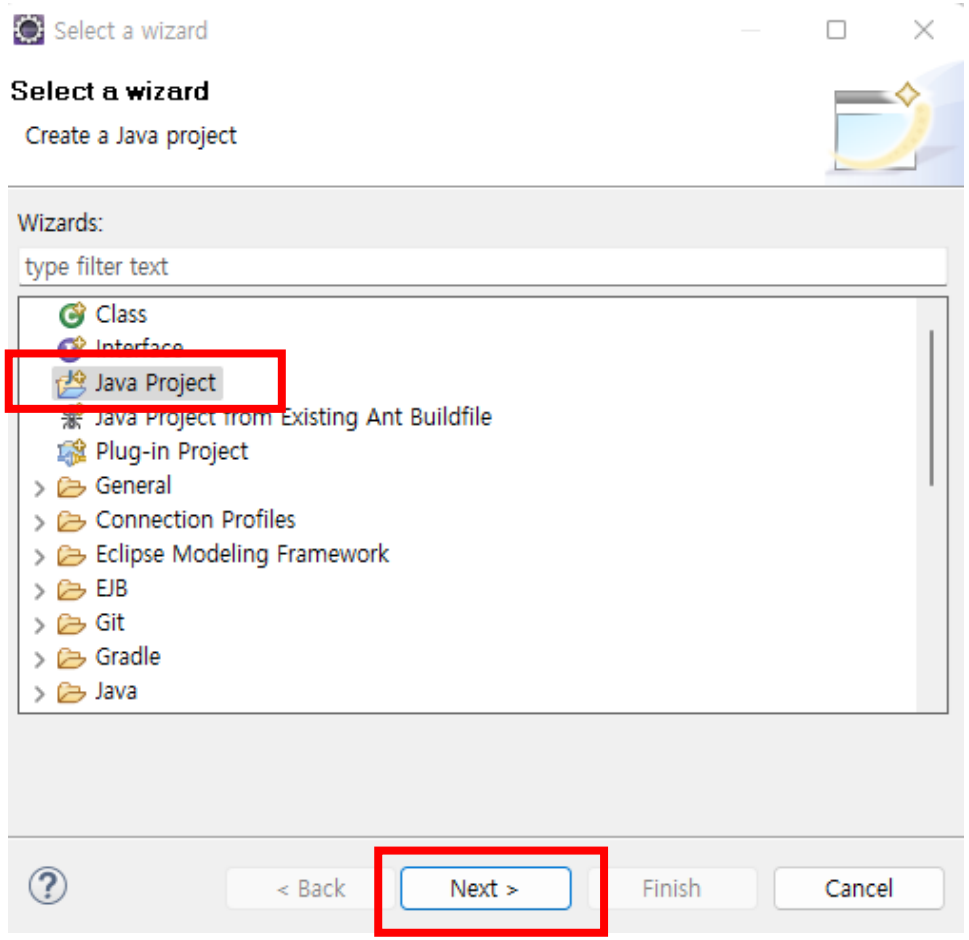
eclipse 설치

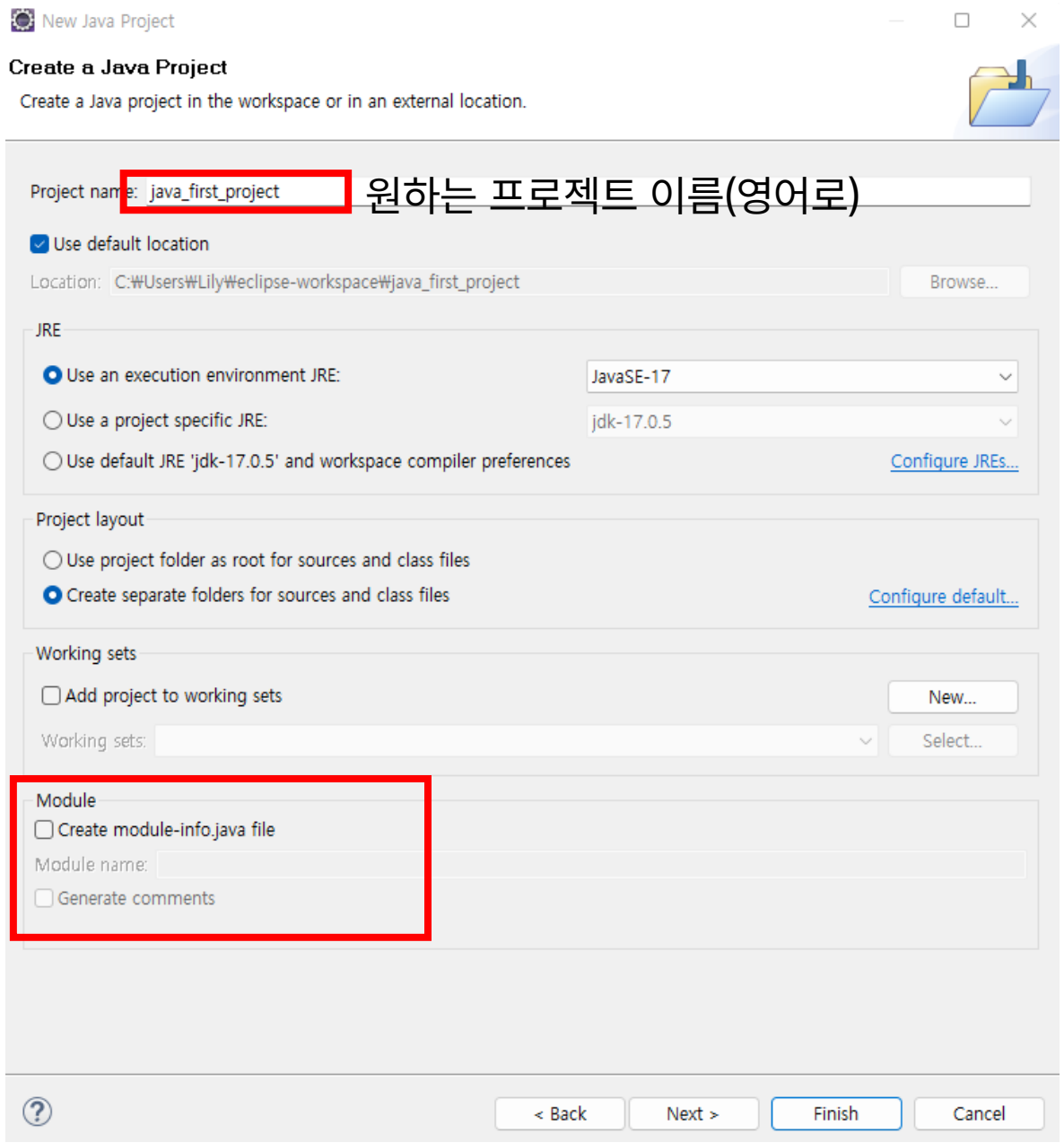
아래와 같이 eclipse IDE가 실행된다면 성공!



프로젝트 만들기

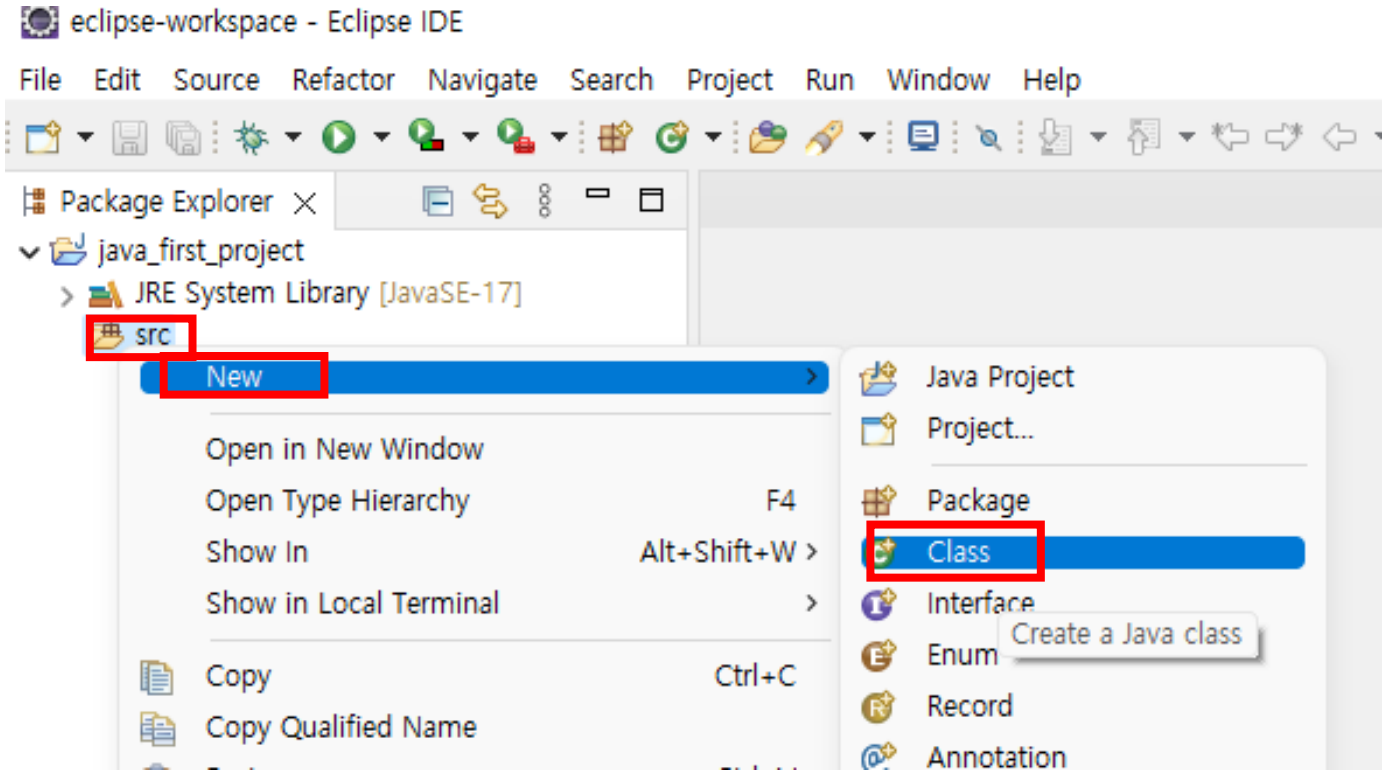

eclipse-workspace - Eclipse IDE
File Edit Navigate Search Project Run Window Help
New Alt+Shift+N
Open File...
Open Projects from File System...
Recent Files
Close Editor Ctrl+W
Close All Editors Ctrl+Shift+W
Save Ctrl+S
Save As...
Save All Ctrl+Shift+S
Revert
Move...
Rename... F2
Refresh F5
Convert Line Delimiters To
Print... Ctrl+P
Import...
Export...
Properties Alt+Enter
Switch Workspace
Maven Project
Enterprise Application Project
Dynamic Web Project
EJB Project
Connector Project
Application Client Project
Static Web Project
JPA Project
Project...
CSS File
JavaScript File
Servlet
Session Bean (EJB 3.x)
Message-Driven Bean (EJB 3.x)
Web Service
Folder
File
JSP File
Example...
Other... Ctrl+N

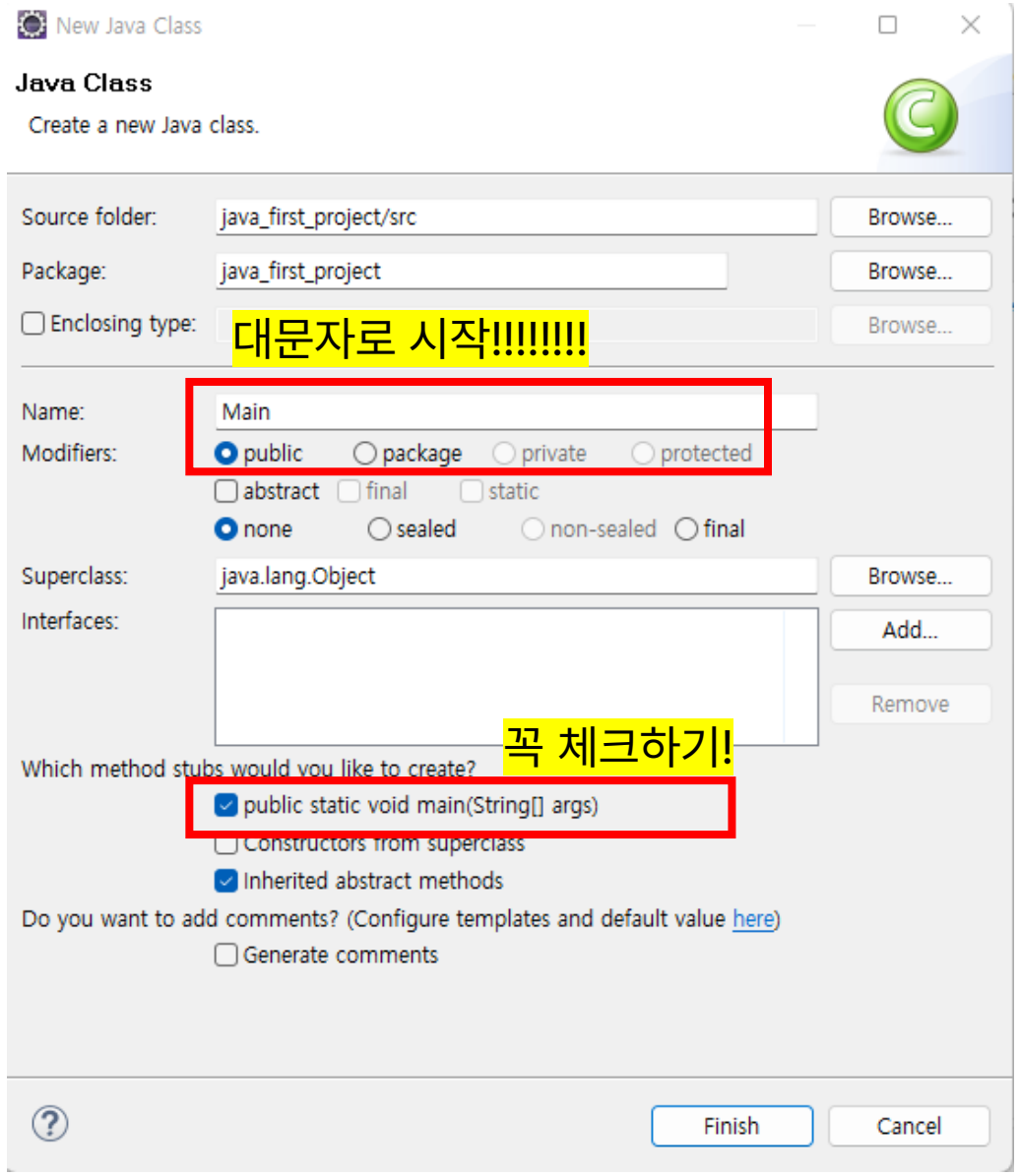

Select a wizard
Select a wizard
Create a Java project
Wizards:
type filter text
Class
Interface
Java Project
Java Project from Existing Ant Buildfile
Plug-in Project
General
Connection Profiles
Eclipse Modeling Framework
EJB
Git
Gradle
Java
< Back
Next >
Finish
Cancel


New Java Project
Create a Java Project
Create a Java project in the workspace or in an external location.
Project name: java_first_project
원하는 프로젝트 이름(영어로)
Use default location
Location: C:₩Users₩Lily₩eclipse-workspace₩java_first_project
Browse...
JRE
Use an execution environment JRE:
JavaSE-17
Use a project specific JRE:
jdk-17.0.5
Use default JRE 'jdk-17.0.5' and workspace compiler preferences
Configure JREs...
Project layout
Use project folder as root for sources and class files
Create separate folders for sources and class files
Configure default...
Working sets
Add project to working sets
New...
Working sets:
Select...
Module
Create module-info.java file
Module name:
Generate comments
< Back
Next >
Finish
Cancel

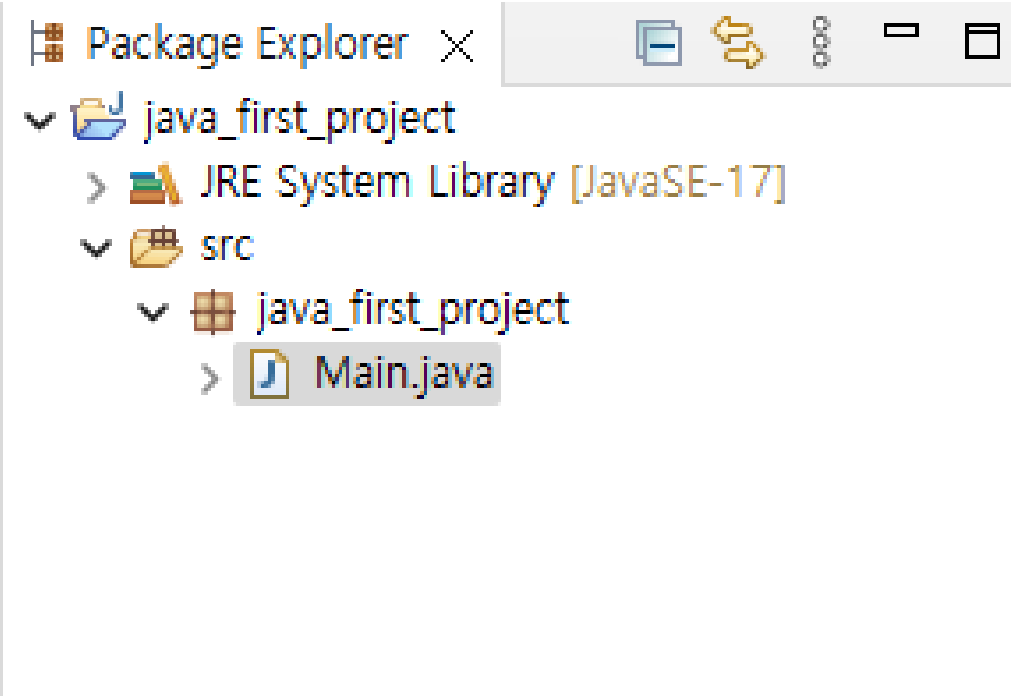
Open Associated Perspective?
Open the Java perspective?
This perspective is designed to support Java development. It offers a Package Explorer, a Type Hierarchy, and Java-specific navigation actions.
Remember my decision
Open Perspective
No

프로젝트 만들기


eclipse-workspace - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
java_first_project
JRE System Library [JavaSE-17]
src
New
Open in New Window
Open Type Hierarchy F4
Show In Alt+Shift+W
Show in Local Terminal
Copy Ctrl+C
Copy Qualified Name
Java Project
Project...
Package
Class
Interface
Enum
Record
Annotation
Create a Java class


New Java Class
Java Class
Create a new Java class.
Source folder: java_first_project/src
Package: java_first_project
Enclosing type:
대문자로 시작!!!!!!!!
Name: Main
Modifiers: public package private protected
abstract final static
none sealed non-sealed final
Superclass: java.lang.Object
Interfaces:
Add...
Remove
Browse...
꼭 체크하기!
Which method stubs would you like to create?
public static void main(String[] args)
Constructors from superclass
Inherited abstract methods
Do you want to add comments? (Configure templates and default value here)
Generate comments
Finish
Cancel

프로젝트 만들기


Package Explorer
java_first_project
JRE System Library [JavaSE-17]
src
java_first_project
Main.java

IntelliJ 설치

IntelliJ

- Java 개발 툴 중 하나
- JetBrains 사에서 제작한 상용 자바 통합 개발 환경
- 간편한 플러그인 설치 (VS Code의 확장 프로그램 같은 것)
- 직관적인 UI

- Eclipse 프로젝트와 구조가 약간 다름
- 유료

다운로드 IntelliJ IDEA

Windows macOS Linux

버전: 2022.2.3
빌드: 222.4345.14
2022년 10월 5일
릴리스 노트 ↗

시스템 요구 사항
설치 안내
기타 버전
타사 소프트웨어

Ultimate

웹 및 엔터프라이즈 개발용

다운로드 .exe

30일 무료 평가 이용 가능

Community

JVM 및 Android 개발용

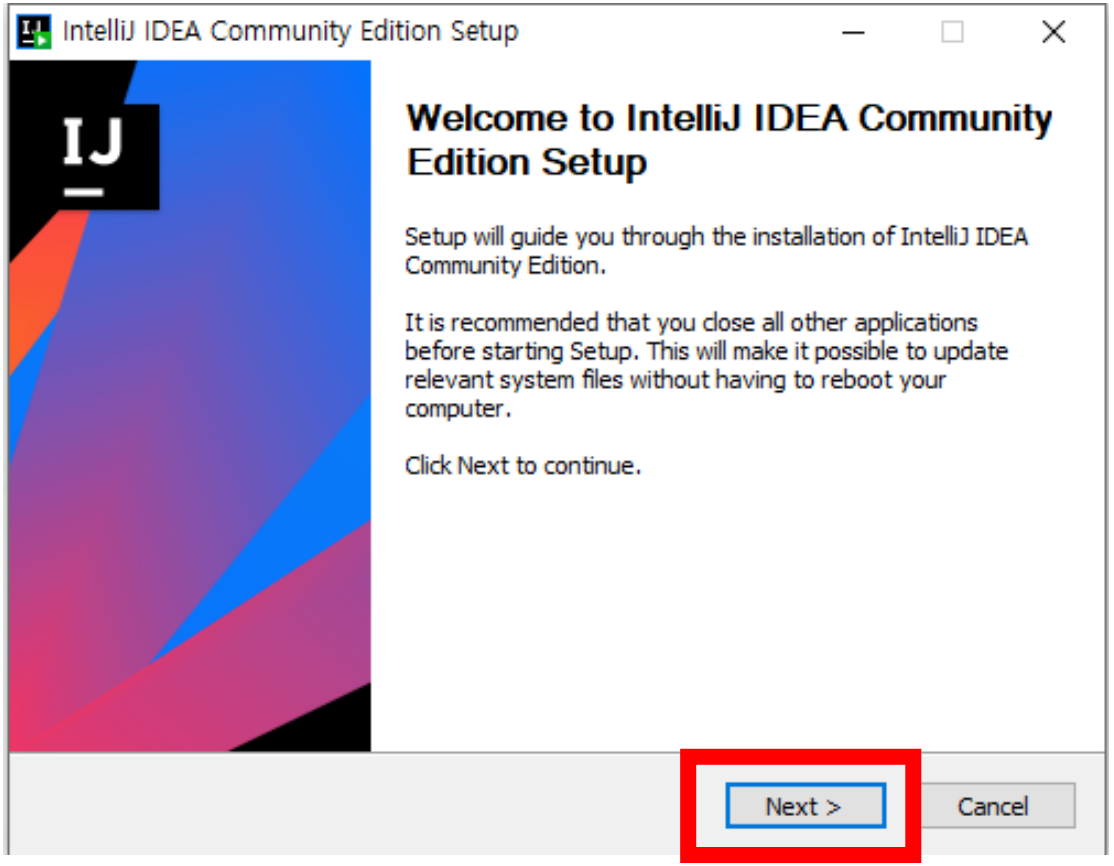
다운로드 .exe

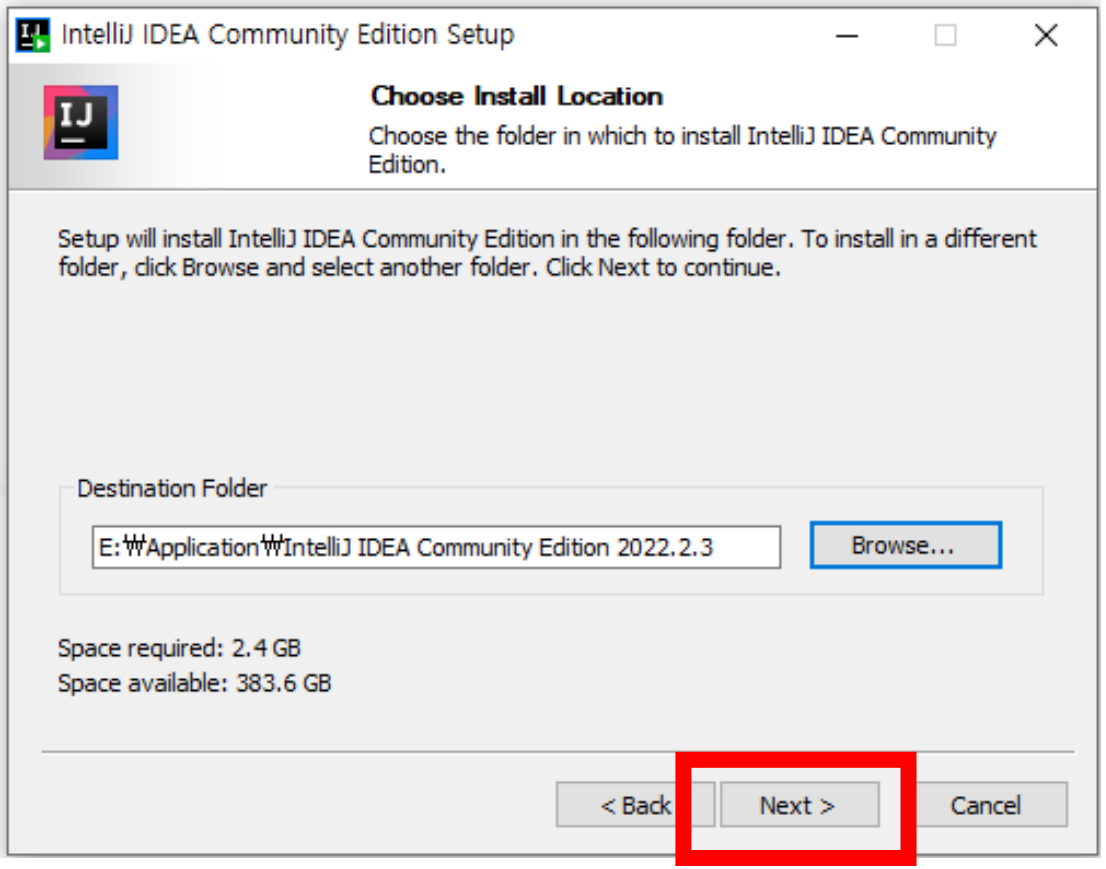
무료, 오픈 소스로 빌드됨

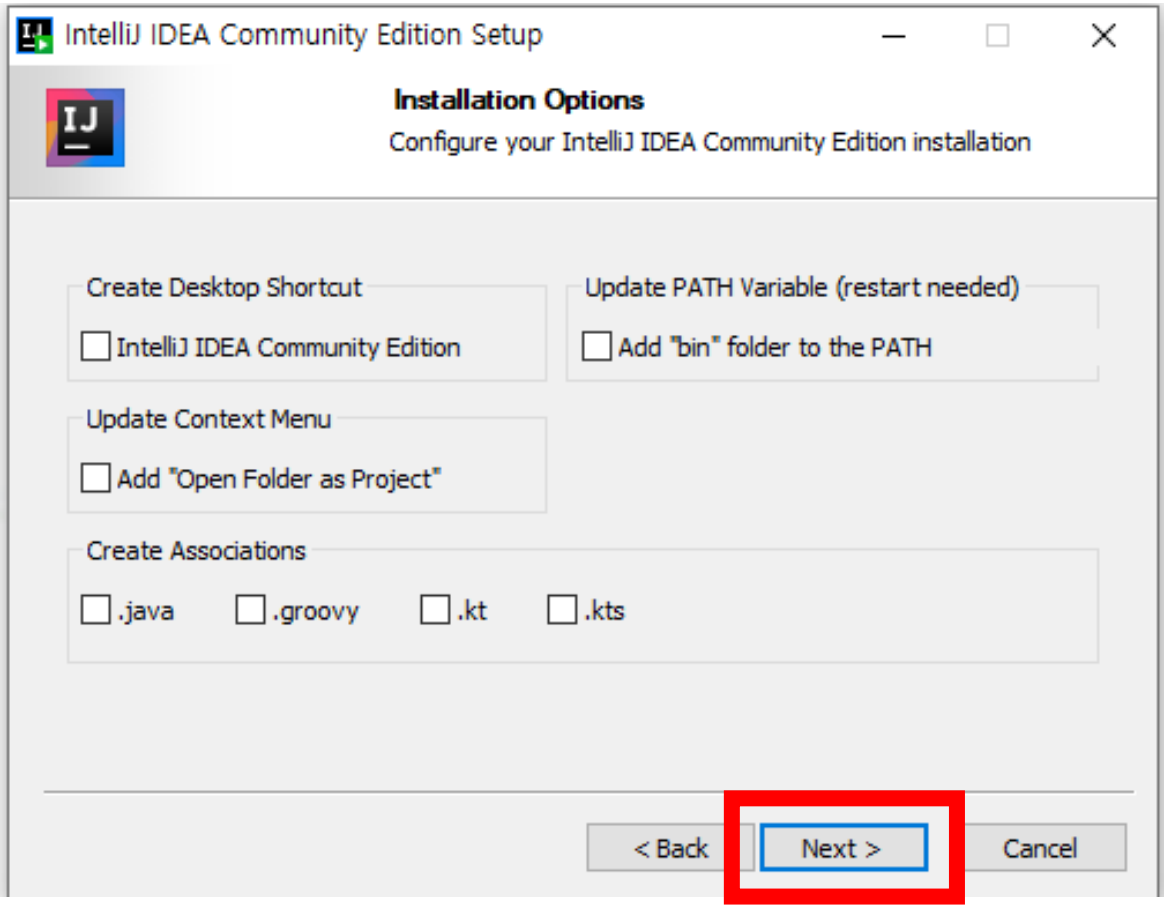
	IntelliJ IDEA Ultimate
Java, Kotlin, Groovy, Scala	✓

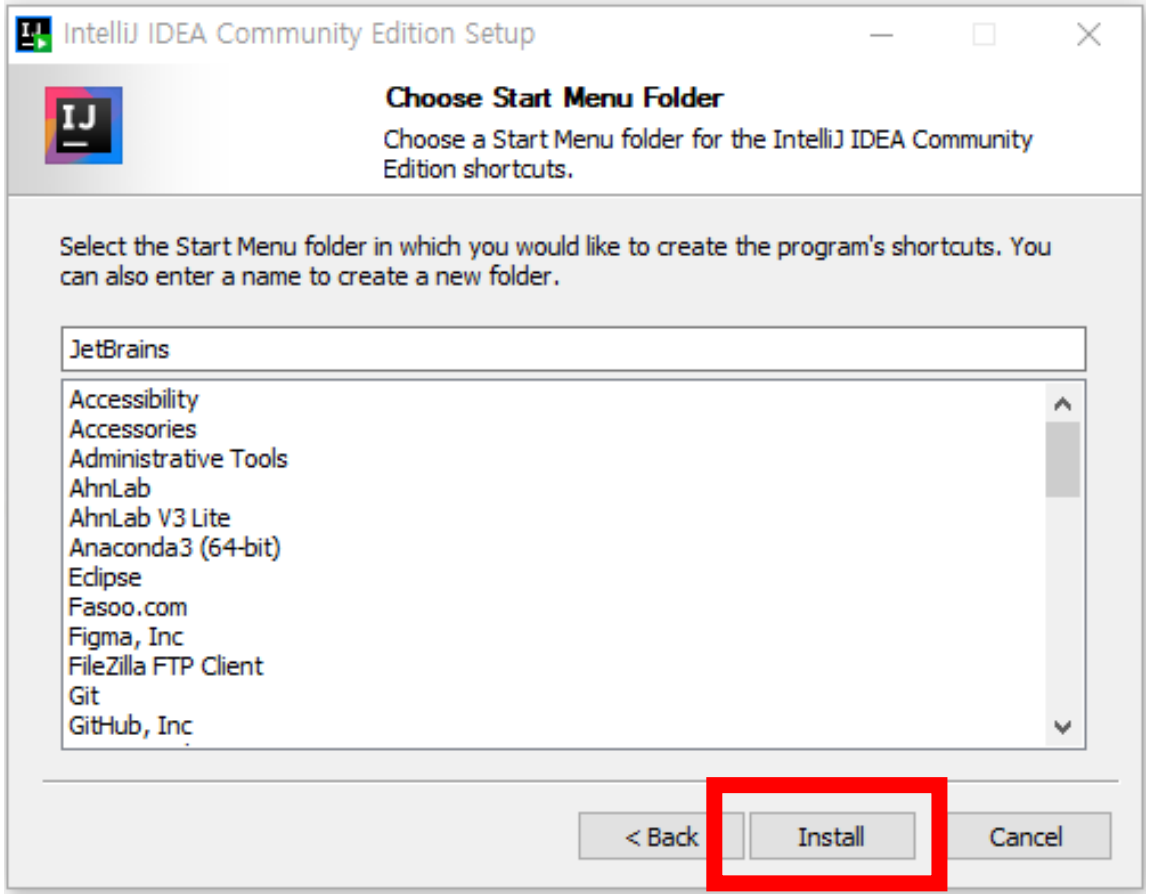
[X]
Cookies and IP
web content and
experience. Our
address for the
JetBrains may
individual st
offers and ad

와다다 Next 눌러서 설치 완료하기


IntelliJ IDEA Community Edition Setup
IJ
Welcome to IntelliJ IDEA Community Edition Setup
Setup will guide you through the installation of IntelliJ IDEA Community Edition.
It is recommended that you close all other applications before starting Setup. This will make it possible to update relevant system files without having to reboot your computer.
Click Next to continue.
Next >
Cancel


IntelliJ IDEA Community Edition Setup
Choose Install Location
Choose the folder in which to install IntelliJ IDEA Community Edition.
Setup will install IntelliJ IDEA Community Edition in the following folder. To install in a different folder, click Browse and select another folder. Click Next to continue.
Destination Folder
E:₩Application₩IntelliJ IDEA Community Edition 2022.2.3
Browse...
Space required: 2.4 GB
Space available: 383.6 GB
< Back
Next >
Cancel

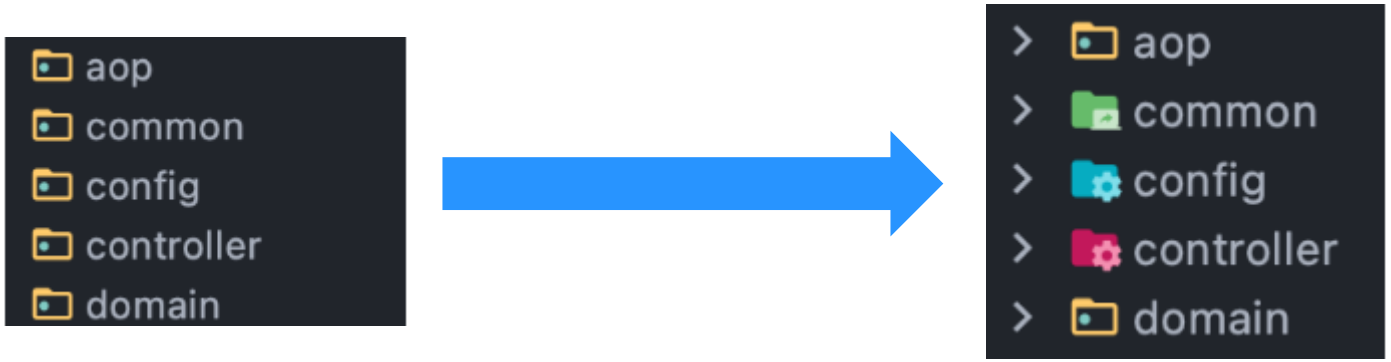

IntelliJ IDEA Community Edition Setup
Installation Options
Configure your IntelliJ IDEA Community Edition installation
Create Desktop Shortcut
IntelliJ IDEA Community Edition
Update PATH Variable (restart needed)
Add "bin" folder to the PATH
Update Context Menu
Add "Open Folder as Project"
Create Associations
.java
.groovy
.kt
.kts
< Back
Next >
Cancel


IntelliJ IDEA Community Edition Setup
Choose Start Menu Folder
Choose a Start Menu folder for the IntelliJ IDEA Community Edition shortcuts.
Select the Start Menu folder in which you would like to create the program's shortcuts. You can also enter a name to create a new folder.
JetBrains
Accessibility
Accessories
Administrative Tools
AhnLab
AhnLab V3 Lite
Anaconda3 (64-bit)
Eclipse
Fasoo.com
Figma, Inc
FileZilla FTP Client
Git
GitHub, Inc
< Back
Install
Cancel

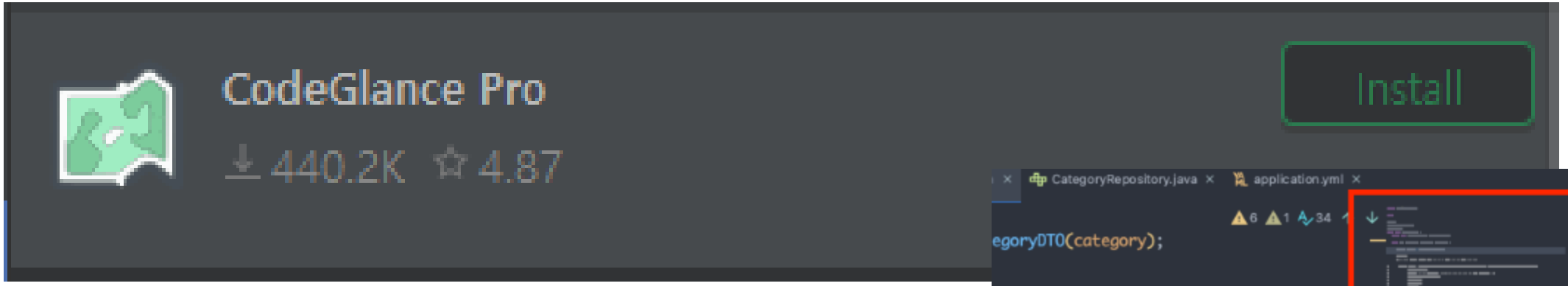
플러그인 추천

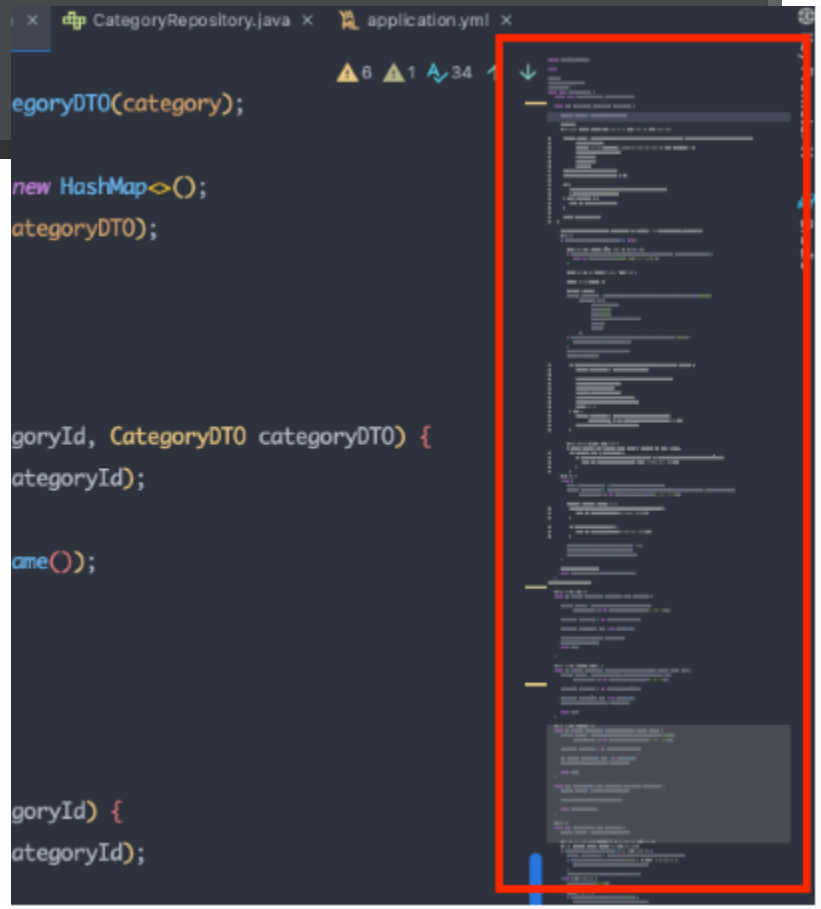
버전 상관 X

- 프로젝트 구조(패키지, 폴더, 파일)을 Atom ide 디자인을 입혀서 아이콘 모양을 바꿔주는 플러그인



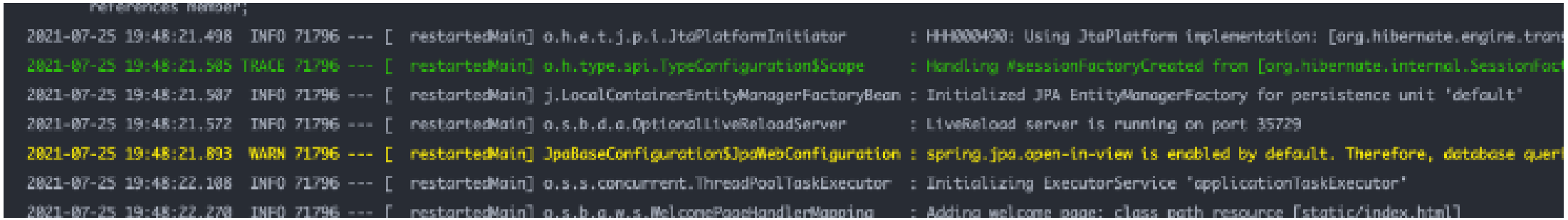
- IntelliJ 가 예뻐진다…;;





- 대략적인 전체적인 코드의 양을 체크하거나 위에 적어 놓은 메소드를 찾을 때 유용하다.

- Console 출력을 상황에 맞게 색상 별로 나눠서 출력해준다.



플러그인 추천

ver. Ultimate

- VSCode에서의 Git Lengs
- 코드별 Git commit 내용 파악이 가능하다.

간단한 코드 실행

자바 소스 파일 기본 구조

Main.java

```
1  public class Main {
2      public static void main(String[] args) {
           System.out.println("Hello, World!");
4      }
5  }
```

클래스

메소드

Java 프로그램은 **main 메소드** 안에서 실행을 시작

자바 소스 파일 기본 구조

Main.java

main 메소드
= 프로그램 실행 진입점(entry point)

```
1  public class Main {
2      public static void main(String[] args) {
           System.out.println("Hello, World!");
4      }
5  }
```

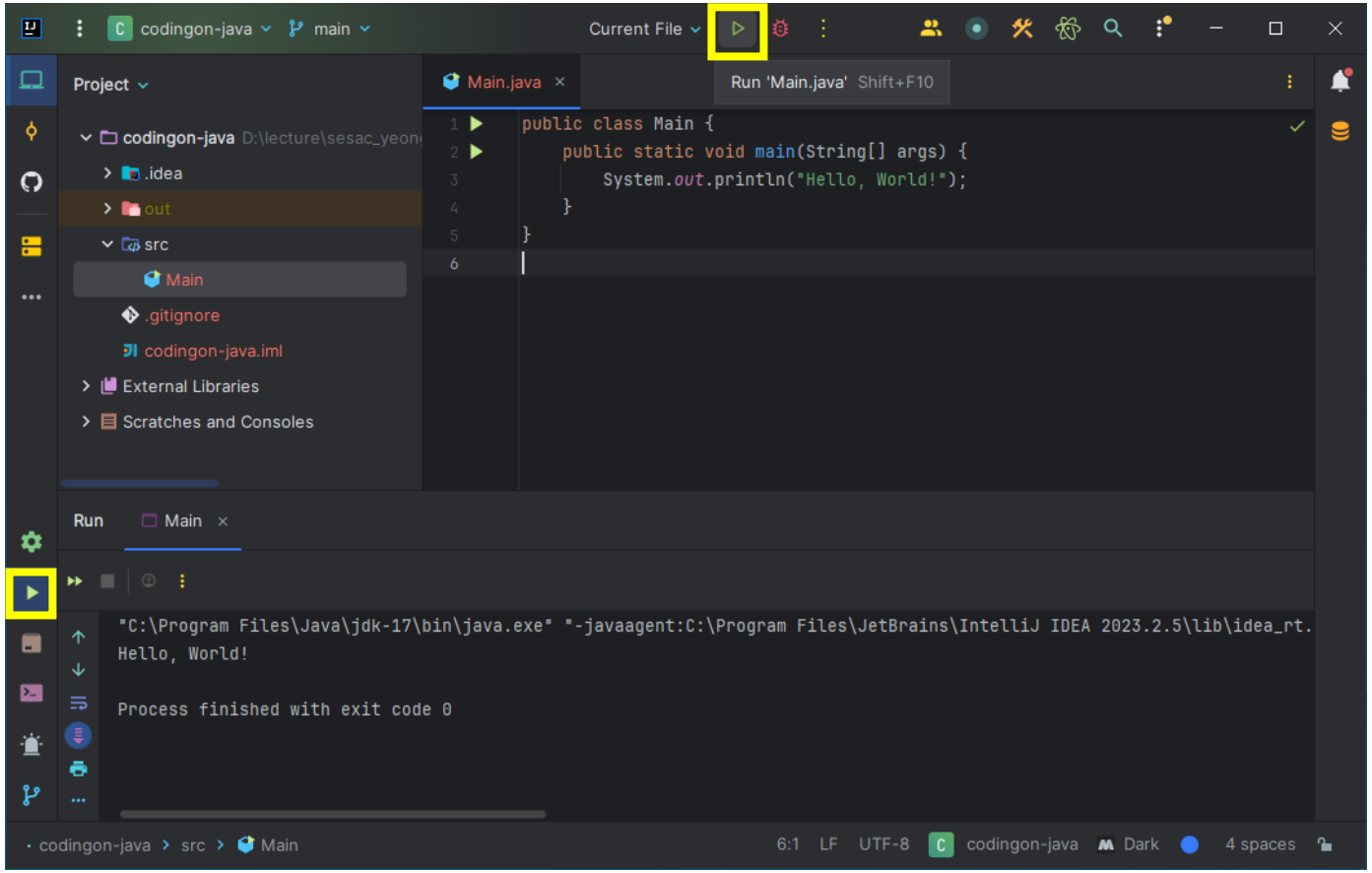
클래스 블록

메소드 블록

괄호 안의 내용을 console view에 출력

main 메소드 안에 **System.out.print("Hello, World!");** 작성해보기
*JavaScript에서의 console.log()와 동일한 역할

코드 실행하기



1. 파일 실행하기
   - Windows: Shift + F10
   - macOS: Ctrl + R

2. 다시 실행하기
   - Windows: Ctrl + F5
   - macOS: Cmd + R

Git 연동

프로젝트에 Git/GitHub 연결하기

- Eclipse 프로젝트에 Git/GitHub 연결하기
  - Git hub(깃허브) :: eclipse(이클립스) 연동방법 및 커밋하기

- IntelliJ 프로젝트에 Git/GitHub 연결하기
  - IntelliJ에서 Git/GitHub 사용하기 - 기존 프로젝트 연동하기